

What Done Looks Like

Every component the finished project should contain, with a spec, an owner, an acceptance criterion, and an honest mark for what you already have. **Sixty-nine components; nine of them exist today.** That sounds worse than it is — the nine you have are the ones that are hardest to start.

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§1

The system at a glance

Six layers. Green is built, amber is built but flawed, red is missing. Everything red is specified in the sections that follow.

2 PARTIAL 7

MISSING 57

3

Have 2 – built and working

Partial 7 – built but flawed

Missing 57 – not started

Stretch 3 – only if ahead



READ THE DIAGRAM THIS WAY

Do not read the red as ninety per cent left to do. Layers 1, 2, 3 and 5 all have something green or amber, which means **you have a working end-to-end pipeline** – image in, prediction out, explanation rendered. That is the hard thing to start and you have already started it. Layers 4 and 6 are almost entirely red, and that is exactly where the marks and the paper live. The work

ahead is less "build the system" than "measure the system", and measurement components are small, independent and fast once the substrate is right.

§2 Data assets

Four datasets, each doing a job no other one can do. Owner: Sai.

ID	ASSET	JOB IT DOES	ACCEPTANCE CRITERION	STATUS
D-1	Kermany chest X-ray 5,856 paediatric, binary	The baseline you already have. Keep it as the "v1, and here is what was wrong with it" narrative.	Re-split patient-level; documented as paediatric in every output	HAVE
D-2	VinDr-CXR 18,000 PA, 22 findings, boxes	The single most important addition. Radiologist bounding boxes are what let you score whether an explanation is correct.	Downloaded, boxes parsed, mIoU computable end to end	MISSING
D-3	CheXpert or NIH ChestX-ray14 multi-label, 8-14 findings	Moves you off saturated binary pneumonia into the realistic clinical setting where explanations actually break.	Multi-label training runs; per-pathology AUROC reported	MISSING
D-4	External validation set a source you did not train on	The most convincing table in your report, and the standard reviewers apply.	Train on one source, test on another; report the drop honestly	MISSING
D-5	Concept labels	Feeds the concept bottleneck. Derive	Concept matrix built; derivation documented in	MISSING

	12–20 radiological concepts	from CheXpert finding labels plus VinDr local labels — do not annotate from scratch.	the paper	
D-6	Study case set 40–60 cases, stratified	The fixed set clinicians see. Includes the ~25% deliberately-wrong subset.	Frozen and version-tagged before the study opens; never regenerated mid-study	MISSING
D-7	Split manifests committed file lists	Makes every number in the report reproducible by anyone who clones the repo.	Patient-disjoint, fixed seed, committed to <code>configs/splits/</code> , enforced by a test	MISSING

START D-2 THIS WEEK

VinDr-CXR is on PhysioNet under credentialed access — you need a completed CITI training course and an approved data use agreement, which takes days to a couple of weeks. It gates \$6's entire localisation block. Request it before you write another line of model code.

§3 Model inventory

Four models, and each exists to answer a specific question. If you cannot say what question a model answers, do not train it. Owner: Sai.

ID	MODEL	QUESTION IT ANSWERS	ACCEPTANCE CRITERION	STATUS
M-1	VGG16 binary baseline	"What does a naive 2020-style pipeline do, and how does it fail?" Your v1, kept deliberately.	Reproducible from config; the border-attribution failure documented, not deleted	PARTIAL
M-2	DenseNet-121	"What is a competent	Per-pathology AUROC	MISSING

	multi-label	modern baseline?" This becomes the backbone every explanation method is measured on.	with bootstrap 95% CIs, internal and external	
M-3	Concept bottleneck	"Can an explanation be faithful by construction, and what does it cost?" Your headline contribution.	Within ~2 AUROC points of M-2; test-time concept intervention works end to end	MISSING
M-4	Patch self-explainable MedicalPatchNet-style	"Does ante-hoc beat post-hoc on localisation?" Also your fallback if M-3 will not converge.	Localisation hit rate beats Grad-CAM on M-2 at comparable AUROC	FALLBACK
M-5	Deployment export ONNX or TorchScript	Makes the demo fast enough that a radiologist does not wait.	Inference under 500 ms per image on CPU	MISSING

Fixes M-1 needs regardless (from the repo audit): matching `preprocess_input` for the VGG16 weights, `GlobalAveragePooling2D` in place of `Flatten`, `class_weight` for the 3:1 imbalance, validation monitoring on AUC rather than accuracy over 16 images, and a tuned decision threshold instead of a fixed 0.5.

§4 Explanation methods

Six post-hoc methods on one shared backbone, plus the two ante-hoc models. All behind one registry so adding a method is a one-line change.
Owner: Sameer.

TD	METHOD	WHY IT IS IN THE SET	STATUS
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X-1	Grad-CAM	The incumbent and the thing you are testing. Must be fixed first: pre-activation logit, predicted class, one shared implementation.	PARTIAL
X-2	Grad-CAM++	Direct successor; handles multiple instances of a pathology better, which the CheXlocalize benchmark shows is where Grad-CAM degrades.	MISSING
X-3	Score-CAM	Gradient-free, so it sidesteps the saturation problem entirely. A useful contrast in your results table.	MISSING
X-4	Integrated Gradients	Axiomatic method with completeness guarantees. Included in the Stanford benchmark, so your numbers are comparable to published ones.	MISSING
X-5	Occlusion	Slow but model-agnostic and conceptually unimpeachable. Effectively a ground-truth-ish reference for the faster methods.	MISSING
X-6	SHAP	Kept for the comparison your proposal promised — but reported honestly, including its cost and its poor fidelity on correlated pixels. Optionally moved to tabular metadata where it is actually appropriate.	PARTIAL
X-7	Concept attribution	The M-3 concept vector itself. Not post-hoc — this is the explanation, and it is faithful by construction.	MISSING
X-8	Patch contribution map	M-4's per-patch scores. Also faithful by construction.	FALLBACK
X-9	Method registry	Name → implementation mapping so the benchmark loops over methods rather than hardcoding them. This is what makes §6 cheap.	MISSING

§5 Uncertainty and abstention

Entirely missing today, cheap to add, and it answers the question a clinician actually asks. Owner: Vedhanshu.

ID	COMPONENT	SPEC	ACCEPTANCE CRITERION	STATUS
U-1	Calibration split	A held-out slice used only for conformal calibration, disjoint from train, val and test.	Documented in the split manifest; never touched during training	MISSING
U-2	Split conformal predictor	Distribution-free prediction sets at a user-set error rate. Roughly thirty lines on top of a trained model.	Empirical coverage on test matches the target within sampling error	MISSING
U-3	Abstention policy	Refer to a radiologist when the prediction set is larger than one label, or when set size exceeds a threshold.	Accuracy-versus-abstention-rate curve plotted across thresholds	MISSING
U-4	Coverage verification	An explicit experiment demonstrating the guarantee holds on your data, not just in theory.	Coverage plot at 3+ target levels; committed as a report figure	MISSING



WHY THIS PUNCHES ABOVE ITS WEIGHT

A softmax confidence of 0.87 means very little to a clinician and nothing statistically. A prediction set carrying a 90% coverage guarantee is a claim you can defend in a viva and quantify in a paper. It is the cheapest genuine differentiator on this page, and it pairs naturally with the concept bottleneck: concepts say *why*, conformal says *how sure*, and the abstention flag says *ask a human*.

§6 The evaluation suite

This is the project. Everything above exists so that this section can produce numbers. Owner: Sameer.

Classification performance

ID	METRIC	WHY NOT JUST ACCURACY	STATUS
E-1	Per-pathology AUROC with bootstrap 95% CIs	A point estimate with no interval is not a result. CIs are what let a reviewer judge whether two models actually differ.	MISSING
E-2	Sensitivity and specificity, reported separately	Your current 85.4% accuracy hides 63.7% specificity. Never report the aggregate alone again.	PARTIAL
E-3	ROC, PR curves, confusion matrix	Already built and working — the strongest file in your repo.	HAVE
E-4	External validation performance	Internal numbers on a single source prove almost nothing about generalisation.	MISSING
E-5	Calibration curve and Brier score	A model can rank well and still report meaningless confidences. Relevant because your UI shows a confidence number.	MISSING



Explanation quality — the part nobody else does

ID	METRIC	WHAT IT ESTABLISHES	STATUS
E-6	Localisation: mIoU and hit rate against VinDr boxes / CheXlocalize masks	Whether the explanation points at the pathology at all. The load-bearing measurement of the whole project.	MISSING
E-7	Faithfulness: deletion and insertion AUC, ROAD	Whether the highlighted region is what the model actually used — separate from whether it is anatomically correct.	MISSING

E-8	Sanity checks: cascading model-parameter and label randomisation	Whether a method produces convincing maps from a randomised model. Methods that fail this are not explaining anything.	MISSING
E-9	Robustness and complexity via the Quantus toolkit	Stability under small perturbations, and how concentrated the explanation is. Off-the-shelf and citable.	MISSING
E-10	Stratification by lesion size and multiplicity	The published finding is that saliency degrades on small, multiple, complex lesions. Confirming or contradicting it is a result either way.	MISSING
E-11	Runtime per explanation	Decides what can ship. SHAP being 100× slower than Grad-CAM is a finding, not an inconvenience.	MISSING

Human evaluation

ID	MEASURE	NOTES	STATUS
E-12	Diagnostic accuracy per condition (C0–C3)	Within-subject, counterbalanced. Valid at n=8 if analysed correctly.	MISSING
E-13	Agreement rate on incorrect predictions	Your headline result. The rate at which a clinician who would have been right unaided follows a wrong AI.	MISSING
E-14	Appropriate reliance, reported as a 2×2	Agrees-when-right and disagrees-when-wrong are different things; a single "trust" number conflates them.	MISSING
E-15	Time per case	Speed without accuracy is automation bias, not efficiency.	MISSING
E-16	System Causability Scale	A validated instrument beats an ad-hoc Likert, and it is citable.	MISSING

§7 Responsible-AI audits

Four audits. You already have visual evidence motivating the first one.
Owner: Sai, with Vedhanshu on the compliance mapping.

ID	AUDIT	METHOD	STATUS
A-1	Shortcut audit	Border-masking test, laterality-marker masking, blurred-image source probe, and attribution-in-lung-field ratio. Your current Grad-CAM output already suggests all four will find something.	MISSING
A-2	Subgroup performance	AUROC broken down by sex, age band and view position wherever metadata allows. Published work shows sensitive attributes are readable from a single radiograph at AUROC ≥ 0.90.	MISSING
A-3	Failure gallery	Your ten worst cases with the explanation shown and a sentence of analysis each. Cheap, and it is the section examiners read most carefully.	MISSING
A-4	EU AI Act mapping	An appendix mapping your design to Articles 12 (logging), 13 (transparency) and 14 (human oversight). Not certification — design evidence. High-risk obligations became applicable 2 August 2026.	MISSING

§8 The clinician interface

Five screens. Red bullets are the ones that carry the project's argument — build those first. Owner: Chandraraj, with Vedhanshu on the API.

Case reader GET /case/:id – the primary screen DICOM viewer: windowing zoom	Concept intervention Interaction within the case reader Toggle or adjust any predicted
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Free view: windowing, zoom, pan

Prediction with the conformal prediction set, not a bare percentage

Concept evidence panel with per-concept confidence bars

Explanation overlay with an opacity slider and a method selector

Abstention banner, visually distinct from a prediction

Method comparison: two explanations side by side

"I disagree" capture with a free-text reason

Case metadata and ground truth, hidden until the clinician commits

Toggle or adjust any predicted concept

Diagnosis and prediction set update live from the edited vector

A visible delta: what the change did to the output

Reset to the model's own concept vector

Interventions written to the audit log

This is the demo moment. No heatmap-only project can show it.

Model dashboard

GET /dashboard

Every number read from

`metrics.json` — never a literal

Run timestamp, config name and model hash beside each figure

Per-pathology AUROC table with confidence intervals

Explanation benchmark matrix: method $\times$ metric

Conformal coverage plot

"Not yet evaluated" state when metrics are absent

Dataset explorer

GET /dataset

Counts computed from disk, not hardcoded

Class distribution and co-occurrence matrix

Split manifest with the patient-disjointness check shown as passing

Sample grid with provenance labels

Explicit note that the source is paediatric

Study mode

GET /study/:participant

Randomised case order, counterbalanced across arms C0–C3

Arm-appropriate display: hides

Audit log

GET /audit — Article 12 evidence

Every inference: input hash, model version, output, explanation shown

Every clinician action: agree,

explanations in C0/C1

Response capture with per-case timing

Blinding: no ground truth and no arm label visible

Progress indicator and resumable sessions

Export to a tidy CSV for analysis

disagree, intervene, abstain

Immutable append-only table with timestamps

CSV export for the report appendix

One database table. Costs an afternoon, buys a compliance section.

PICK ONE FRONTEND

You currently have a partial Flask app and a partial Lovable/React app, and neither runs. Two half-built frontends doubles the maintenance surface and you will be asked in the viva why there are two. Finish the Flask templates now to unblock the science, and rebuild in React during Phase 4 only if the API contract has stabilised and Chandraraj has the time.

§9 Engineering substrate

Unglamorous, and the reason everything else can move fast. Owner: Vedhanshu.

ID	COMPONENT	ACCEPTANCE CRITERION	STATUS
G-1	Real git history	Pushed from a working tree, not the web uploader. Meaningful commits, one per change.	MISSING
G-2	<code>.gitignore</code>	Dataset, weights and uploads excluded. The 43 committed test images removed from the working tree.	MISSING
G-3	Installable package <code>src/xai_cxr/</code>	App and scripts both import from it. Zero duplicated functions.	MISSING
G-4	YAML configs	No hyperparameter, path or seed appears as a literal in code.	MISSING

G-5	<code>pyproject.toml</code> with pins	Fresh clone installs and runs on a declared Python version.	PARTIAL
G-6	Makefile	<code>setup / train / eval / explain / audit / serve / test</code> all work.	MISSING
G-7	Test suite	Splits disjoint, Grad-CAM shape and class-correctness, metrics schema, all routes 200.	MISSING
G-8	Experiment tracking	Every run logged with its config. A results table anyone on the team can read.	MISSING
G-9	Docker	Clone to running demo in two commands. Non-negotiable before the viva.	MISSING
G-10	<code>CLAUDE.md</code>	Encodes the hard rules so a coding agent cannot reintroduce the defects you just fixed.	MISSING
G-11	CI on push	Tests run on every push. Catches the broken-repo problem automatically.	STRETCH

§10 Documents and artefacts

What you hand over in May, plus what you file along the way. Owner: Vedhanshu on documentation, Sameer on the paper and ethics.

ID	ARTEFACT	NOTES	DUE	STATUS
P-1	IEC ethics application	Protocol, participant information sheet, consent form, case-set description, data-handling statement. The single most time-critical item in the project.	Sept–Oct 2026	MISSING
P-2	Rewritten literature	10–15 rows, covering post-hoc critiques, self-explanable	Oct 2026	PARTIAL

	literature review	critiques, self-explainable architectures, evaluation frameworks and human-factors work — not five rows anchored on 2017–19.		
P-3	Paper I — the benchmark	Localisation and faithfulness of post-hoc vs self-explainable methods. No ethics dependency, so it can be submitted while recruitment runs.	Mar 2027	MISSING
P-4	Paper II — the human study	Effect of concept explanations and abstention on agreement with incorrect predictions. Extended or separate.	Jun–Aug 2027	MISSING
P-5	arXiv preprint	Free, immediate, citable. Post at submission; it timestamps the work regardless of the venue decision.	Mar 2027	MISSING
P-6	Final project report	With a genuine limitations chapter — paediatric source, derived concept labels, small clinician sample, retrospective data, no prospective validation.	May 2027	MISSING
P-7	Model card	Intended use, out-of-scope use, training data, performance by subgroup, known failure modes.	Apr 2027	MISSING
P-8	CLAIM 2024 checklist	Completed and attached as an appendix. Free marks, and filling it in surfaces what you forgot.	Apr 2027	MISSING
P-9	Dataset datasheet	Provenance, label origin, known biases, licence for every dataset you touch.	Dec 2026	MISSING
P-10	Decisions log	One entry per real choice, written as you go. In April this file writes your methodology justifications	Continuous	MISSING

		your methodology, assumptions and your viva answers.		
P-11	Public code release	Repo with weights, benchmark scripts, and one-command reproduction of the headline table.	May 2027	PARTIAL
P-12	Software copyright registration	Realistic IP outcome, replacing the patent deliverable. See the project review on Section 3(k).	Apr 2027	MISSING
P-13	Demo video, 5 minutes	Examiners watch it, and so do reviewers when you link it from the paper.	May 2027	MISSING

§11 Repository manifest

The finished tree. Every path here is referenced by a component ID above.